

NextFlex Small RNA Library V4

Prepared by:	Robert Cormier	2023-03-27
Revised by:	Brandon Hannay	2023-07-21
Approved by:		

Version	Enforcement Date
2	

REVISIONS FROM THE ORIGINAL VERSION

Version	Summary of modifications	Date and initials
2	Reduced clutter in protocol.	RC 2023-04-19
3	Format	DM 2026-02-25

Date:	
Project:	
Person in charge	
Lib prepared by:	

Goal: This protocol uses the following kit from NextFlex small RNA kit to convert RNA transcripts expressed in cells or tissue into representative cDNA libraries for sequencing using Illumina Novaseq Platform

Materials required:

Item	Source
NextFlex small RNA V4 kit	
Isopropanol	
80% EtOH	
96 well PCR Plate Non-skirted	(Phenix Research, Cat # MPS-499) or similar)
Thin-wall nuclease-free PCR tubes	
Thermocycler	
Heat block	
Magnetic Stand -	96 (Ambion, Cat # AM10027) or similar

NextFlex Small RNA Library V4

NOTE: Remove enzymes from -20 immediately before adding to master mix, flick to mix + 600 x g quickspin, and immediately put back.

NOTE: Vortex and centrifuge reagents (except enzymes) in a microcentrifuge at >12,000 x g for 5 seconds before use to ensure material has not lodged in the can or side of the tube.

NOTE: When mixing reactions by pipetting using the multichannel, increase the total volume on the pipette to ensure adequate mixing. Undermixing will result in failure of the library prep.

NOTE: For all Clean-up steps, pre-aliquote reagents in PCR Strips (Adaptor Depletion Solution, Resuspension Buffer, etc.) to allow use of multichannel.

NOTE: Do not freeze NEXTFLEX Cleanup Beads. Beads should be stored at 4°C and brought to room temperature before use.

RNA Input Recommendations

Total RNA – 1ng -2µg (For high quality RNA)

Purified small RNA - 1ng -2µg (from 1-10µg total RNA) (For high quality RNA)

Synthetic miRNA pool - ≥100pg

If using tRNA/YRNA blockers, 4µl of RNA input is recommended instead of 5µL.

PRE-PCR ROOM

Step A: 3' Adenylated Adaptor Ligation to RNA template: (30 min, 1.5h total)

Required Reagents:

- NEXTFLEX 3' Adenylated Adapter (Red cap)
- NEXTFLEX 3' Ligation Buffer (Red cap)
- NEXTFLEX 3' Ligation Enzyme Mix (Red cap)
- NEXTFLEX tRNA/YRNA Blockers (Red cap, if applicable)

NOTE: Ensure to mix the ligation reaction well. Reagents are viscous and there is little chance of overmixing. Failure to mix adequately will result in failure of library prep.

Reaction set up:

- ☐ Determine positive control (Human Placenta for small RNA, -80°C; Human brain total RNA for total RNA, Ambion AM7962) and Qia spike (dilutions and volumes, -80°C).
- ☐ Dilute positive control and Qia spike with NF H₂O, if applicable.
- ☐ Dilute the RNA samples with NF H₂O to a final volume of 4.5 µl (3.5 µl if using tRNA/YRNA blockers) in a PCR plate and keep on ice.
- ☐ Speedvac RNA samples if applicable then transfer to PCR plate.
- ☐ Add Spike in, ensuring the volume is included in the 5 µL (4 µL if using tRNA/YRNA blockers).
- ☐ Preheat the thermocycler (3' Adaptor Ligation)

- ☐ For poor quality RNA samples (EV RNA), dilute 3' Adenylated adapter ¼ with nuclease-free water to help reduce adapter dimer formation. Determine end volume ((n+2)*vol in mastermix below)
- ☐ Prepare a master mix using the following reagents on ice according to the following table and mix by pipetting: (*RNA + NF water input is 4 µL if using tRNA/YRNA blockers)

Reagent (Lib Prep Kit, -20°C)	Volume/ 1 sample	*X	
RNA input	$x \mu\text{L}$	--	☒
Nuclease-free water	$(5-x) \mu\text{L}$		
NEXTFLEX tRNA/YRNA blockers (optional)	1 µL		
NextFlex 3' Adenylated adaptor (refer to dilution; undiluted for controls)	1 µl		☐
NEXTFLEX 3' Ligation Buffer (viscous)	12.5 µL		☐
NEXTFLEX 3' Ligation Enzyme Mix	1.5 µL		☐
Total	15 µL /well of MasterMix; 20 µL total		

- ☐ Add 15 µl of master mix to each well containing the RNA input and pipette to mix.
- ☐ Place the tubes into the preheated thermal cycler, perform the following cycling conditions: (3' Adapter Ligation)

Temperature	Duration	
25°C	1 hours	☒
4°C	Min. 5 minutes	☒

Overnight incubation (16h) at 16°C might be preferred for plant samples with 2'O-methylated small RNAs. **Do not exceed 16 hours in the thermocycler.**

- ☐ Take out reagents for Step B: Excess 3' Adapter Removal near the end of this incubation.
- ☐ Take out reagents for Step C (except enzyme) near the end of this incubation.
- ☐ Proceed immediately to Step B: Excess 3' Adapter Removal

Step B: Excess 3' Adapter Removal

Required Reagents:

- NEXTFLEX Adapter Inactivation Buffer (Pink cap)

☐ Preheat thermal cycler (Excess 3' Adapter Removal).

For each sample, combine the following reagents on ice:

Reagent (Lib Prep Kit, -20°C)	Volume/ 1 sample	*X	
3' Adapter ligated RNA (from Step A)	20 µl	--	☒
NEXTFLEX Adaptor Inactivation Mix	4 µl		☐
Total	4 µl /well mastermix; 24 µl Total		

☐ Add 4 µl of the Adaptor Inactivation Mix into each sample well and pipette to mix.

☐ Incubate the reactions following the conditions below: (Excess 3' Adapter Removal)

Temperature	Duration	
70°C	2 minutes	☒
4°C	5 minutes	☒
4°C	∞	☒

☐ Proceed immediately to Step C: NEXTFLEX 5' Adaptor Ligation

Step C: NEXTFLEX 5' Adaptor Ligation

Required Reagents

- NEXTFLEX 5' Adaptor (Light purple cap)
- NEXTFLEX 5' Ligation Buffer (Light purple cap)
- NEXTFLEX Ligation Enzyme Mix (Light purple cap)

NOTE: Ensure to mix the ligation reaction until visibly homogenous by vortexing. Reagents are viscous and there is little chance of overmixing. Failure to mix adequately will result in failure of library prep.

- ☐ Dilute the 5' Adaptor in NF water (in a 0.2ml PCR tube) using the same dilution as the 3' Adaptor. Determine end volume ((n+2)*vol in mastermix below)
- ☐ Preheat thermocycler (5' Adaptor Ligation)

- ☐ Prepare the following reagents on ice (except the RNA) and mix by pipetting until visibly homogenous.

Reagent (Lib prep Kit, -20°C)	Volume/ 1 sample	*x	
NEXTFLEX 3' Adaptor Ligated RNA (From Step B)	24µl	--	☒
NEXTFLEX 5' Adapter (Refer to dilution table)	1 µl		☐
NEXTFLEX 5' Ligation Buffer	3 µl		☐
NEXTFLEX 5' Ligation Enzyme Mix	2 µl		☐
Total	6 uL/well mastermix; 30 µl Total		

- ☐ Add 6 ul of the master mix to the 3' Adapter Ligated RNA into each sample well and pipette to mix.
- ☐ Incubate the reactions following the conditions below: (5' Adapter Ligation)

Temperature	Duration	
20°C	60 minutes	☒
4°C	5 minutes	☒
4°C	∞	☒

- ☐ Take out reagents for Step D (except enzyme) near the end of this incubation.
- ☐ Proceed immediately to Step D: Reverse Transcription - First Strand Synthesis.

Step D: Reverse Transcription-First Strand Synthesis

Required Reagents

- NEXTFLEX RT Buffer (Blue cap)
 - NEXTFLEX RT Enzyme Mix (Blue cap)
 - NEXTFLEX RT Primer (Blue cap)
- ☐ Prepare the following reagents on ice (except the RNA) and mix by pipetting until visibly homogenous.
- ☐ Preheat thermal cycler: (First Strand Synthesis).

Reagent (Lib Prep Kit, -20°C)	Volume/ 1 sample	*X	
5' and 3' Adaptor Ligated RNA (From Step C)	30 µl	--	☒
NEXTFLEX RT Primer	1 µl		
NEXTFLEX RT Buffer	7 µl		☐
NEXTFLEX RT Enzyme Mix	2 µl		☐
Total	10uL/well mastermix; 40uL Total		

- ☐ Add 10 µl of the master mix to each RNA sample and pipette to mix.
- ☐ Incubate as follows: “First Strand Synth”

Temperature	Duration	
50°C	60 minutes	☒
90°C	5 minutes	☒
4°C	5 minutes	☒
4°C	∞	☒

***STOPPING POINT: It is encouraged to proceed to Step E: Cleanup. However, samples may be stored at 4°C overnight, with minimal impact on data quality. If proceeding:**

- ☐ Take the cleanup beads from the fridge 30 minutes before the end of incubation.
- ☐ Prepare 80% EtOH (400 µl/sample), isopropanol (90 µl/sample), get out Adaptor Depletion Solution (grey bag – room temp – pre-room), aliquot in PCR strip.

Step E: Bead Cleanup

Required Reagents

- NEXTFLEX Adapter Depletion Solution (Red Cap)
- NEXTFLEX Cleanup Beads (Clear Cap Bottle)
- Isopropanol (User supplied)
- Freshly prepared 80% ethanol (User supplied)
- Nuclease-free water

Cleanup steps on next page →

Step E: Bead Cleanup

- ☐ Transfer samples in a purification plate (40µl/sample). If purification plate is not available, do clean-up in PCR plate
- ☐ Add 40 µL of NEXTFLEX Adapter Depletion Solution and mix well by pipette.
- ☐ Add 40 µL of NEXTFLEX Cleanup Beads and mix well by pipette.
- ☐ Add 90 µL of isopropanol and mix well by pipette.
- ☐ Incubate samples for 5 minutes at RT.
- ☐ Incubate the sample on the magnet for 5 minutes or until solution is clear.
- ☐ Remove and discard the supernatant.
- ☐ Add 200 µL of freshly prepared 80% EtOH, incubate for 30 seconds, and remove ALL the supernatant. ***Important: Do not incubate pellet in ethanol for extended periods. ***
- ☐ Repeat the previous step. (Total : 2 wash)
- ☐ Air-dry sample for 3 minutes on the magnet. After one minute, remove ALL residual liquid that may have collected at the bottom of the well.
- ☐ **Remove plate from magnetic stand** and resuspend bead pellet in 18 µL NF H₂O by pipetting volume up and down. Ensure that beads are completely resuspended.
- ☐ Incubate for 2 minutes.
- ☐ Incubate the sample on the magnet for 3 minutes or until solution is clear.
- ☐ Transfer 16 µL (2 x 8 µl) of supernatant to a new PCR plate.
- ☐ Proceed immediately to Step F: PCR Amplification.

***STOPPING POINT: Place RT product in -20 overnight.**

POST-PCR ROOM

Step F: PCR Amplification

Required Reagents

- PLATE: NEXTFLEX UDI Barcoded Primer Mix
- NEXTFLEX Small RNA PCR Master Mix (Green Cap)

☐ Thaw the index plate primers (BC Plate, -20°C) and place the sample plate, index plate and PCR Master Mix (Lib Prep Kit, -20°C) on ice.

Depending on initial RNA input, select the proper amount of PCR cycles:

Input Amount	PCR cycles
1 µg	12-14
100 ng	15-17
10 ng	18-20
1 ng	22-24*
Plasma/Serum	22-24*
miRNA Control	15-17

*When diluting adapters by 1/4, use 24 PCR cycles.

☐ Pre-heat the cyclor to 80 degree.

☐ **On ice**, add 6 µl of the Small RNA PCR Master Mix to each reaction tube with 16 µl of FS product

☐ Shake the index plate at 1800 x g for 1 minute and quickspin. Add 4 µl of a **different** Index Barcode to each reaction tube **Important:** Note which barcodes are in which samples

Reagent	Volume/ 1 sample	*X	
Purified FS product	16 µl	--	☒
NEXTFLEX® Index Barcode Primer	4 µl	--	☐
NEXTFLEX® Small RNA PCR Master Mix	6 µl	--	☐
Total	25 µl		

☐ Spin the sample down and mix well by pipetting up and down (total vol 25uL). Spin again

☐ Place the tubes into a thermocycler and, with a heated lid, perform the following cycling conditions depending on the initial input and origin of the sample:

Temperature	Duration	
98°C	30 seconds	☒
98°C	10 seconds	☐ 12-24 *number of cycles varies based on the original RNA input amount
65°C	20 seconds	
72°C	15 seconds	
72°C	2 minutes	☒
4°C	∞	☒

- ☐ Take out NEXTFLEX® Cleanup Beads and bring to room temp (30 min)
- ☐ Prepare 80% EtOH, get out Resuspension Buffer

***STOPPING POINT: Place PCR product in -20 overnight.**

Step G-1: Size selection & clean up (Small RNA samples)

***Selects small RNA libraries between 150-160bp**

- ☐ Take the D1000 screen tape and reagents out and allow to sit at room temperature for 30 minutes before use
- ☐ Prepare 80% EtOH (400 µl/sample), get out Resuspension Buffer
- ☐ Ensure that the volume of samples is 26 µL. If less, add NF H₂O to bring the entire volume up to 26 µL.
- ☐ Transfer samples in a purification plate (26 µL/sample) ○ If purification plate is not available, do clean-up in PCR plate
- ☐ Add 34 µL of NEXTFLEX® Cleanup Beads and mix well by pipetting.
- ☐ Incubate for 5 minutes.
- ☐ Incubate the sample on the magnet for 5 minutes or until solution is clear.
- ☐ **DO NOT DISCARD SUPERNATANT.** Transfer 56 µL of supernatant to a new well. This solution contains the amplified product. Take care to not transfer beads along with clear supernatant.
- ☐ Remove plate from magnetic stand.
- ☐ Add 13 µL of NEXTFLEX® Cleanup Beads to each sample and mix well by pipette.
- ☐ Incubate for 5 minutes.
- ☐ Incubate the sample on the magnet for 5 minutes or until solution is clear.
- ☐ Remove and discard supernatant
- ☐ Add 200 µL of freshly prepared 80% EtOH, incubate for 30 seconds, and remove ALL of the supernatant.
- ☐ Repeat the previous step. (Total : 2 wash)
- ☐ Air-dry sample for 3 minutes. After one minute, remove ALL residual liquid that may have collected at the bottom of the well.
- ☐ **Remove plate from magnetic stand** and resuspend bead pellet in 17 µL of Resuspension buffer by pipetting volume up and down. Ensure that beads are completely resuspended.
- ☐ Incubate for 2 minutes
- ☐ Incubate the sample on the magnet for 3 minutes or until solution is clear.
- ☐ Transfer 15 µL (2 x 7.5 µl) of supernatant to a new well of a PCR plate. This is your sequencing library.
- ☐ Seal the plate, shake at 1800rpm for 1 min then quickspin
- ☐ QC the Final Library:
 - Check the size distribution of the final library by Tapestation D1000 Assay (Agilent). Desired peak size is in between 150-160bp.

Note: See Figure 4A in Appendix A for sample Bioanalyzer trace. If significant high molecular weight; greater than 180bp products remain, bring total sample volume to 25 µL with NF H₂O and repeat Step 8.

Even if library fails, adapter-dimer should be observed when using ≥ 18 cycles of PCR. Absence of adapter-dimer (flat tapestation profile) in such cases may indicate undermixing of reaction.

Step G-2: No Size selection and clean-up (Exosomes Samples)

- ☐ Take the D1000 screen tape and reagents out as well as the Qubit HS dsDNA kit and allow to sit at room temperature for 30 minutes before use.
- ☐ Prepare fresh 80% EtOH (400 µl/sample)
- ☐ Retrieve Resuspension Buffer
- ☐ Ensure that the volume of samples is 26 µl. If less, add NF H₂O to bring the entire volume up to 26 µl.
- ☐ Transfer samples in a purification plate (26 µl/sample) ○ If purification plate is not available, do clean-up in PCR plate
- ☐ Add 47 µL of NEXTFLEX Cleanup Beads to each sample and mix well by pipette.
- ☐ Incubate for 5 minutes
- ☐ Incubate the sample on the magnet for 6 minutes or until solution is clear.
- ☐ Remove and discard supernatant
- ☐ Add 200 µL of freshly prepared 80% EtOH, incubate for 30 seconds, and remove ALL of the supernatant.
- ☐ Repeat the previous step. (Total : 2 wash)
- ☐ Air-dry sample for 3 minutes. After one minute, remove ALL residual liquid that may have collected at the bottom of the well.
- ☐ **Remove plate from magnetic stand** and resuspend bead pellet in 17 µL of Resuspension buffer by pipetting volume up and down. Ensure that beads are completely resuspended.
- ☐ Incubate for 2 minutes
- ☐ Incubate the sample on the magnet for 3 minutes or until solution is clear.
- ☐ Transfer 15 µL (2 x 7.5 µl) of supernatant to a new well of a PCR plate. This is your sequencing library.
- ☐ Seal the plate, shake at 1800rpm for 1 min then quickspin
- ☐ QC the Final Library:
 - Check the size distribution of the final library by TapeStation D1000 Assay (Agilent). Desired peak size is in between 150-200bp.

NOTE: Even if library fails, adapter-dimer should be observed when using ≥ 18 cycles of PCR. Absence of adapter-dimer (flat tapestation profile) in such cases may indicate undermixing of reaction.